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NSP/004/112 (OHI 12) Guidance for the Inspection and testing of Wood & Steel Poles

1. Purpose

The purpose of this document is to provide guidance on the inspection, testing, recording and classification of wood & steel poles for use on the CE Electric UK Distribution System.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
OHI 12	June 1999	Inspection, Testing, Recording & Classification Of Wood & Steel Poles In Position
RTN/001/500/701/002	June 1995	GN 70-02 - Guidance Note precautions before Ascending a Wood Pole
RTN/001/500/701/004	Feb 1997	GN 70-04 - Purl Testing Wood Poles

2. Scope

This document includes details on the inspection, testing, recording and classification of wood & steel poles. In addition it provides guidance on the precautions to be taken prior to ascending wood poles.

3. Technical Specification

3.1 Testing of wood poles before climbing

Before a pole is climbed or relied upon for personal support, it is the duty of the person who is to climb the pole to verify that the pole is in a safe condition to climb. Sound external appearance does not necessarily mean that a pole is safe. Poles showing any signs of damage, decay or weakness as identified in this guidance note shall not normally be relied upon for personal support.

The following tests shall be applied to all poles before they are climbed:

- (a) Visual Examination
- (b) The HAMMER test, which consists of striking the pole a sharp blow.
- (c) The PRODDING test, which consists of prodding or probing the surface with the point of a pocket knife or other similar tool.

The condition of the pole shall be classified as follows, according to the symptoms indicated:

Condition	Test	Symptom
Safe	Visual	No damage
	Hammer	A good ring
	Prodding	No decay
Unsafe	Visual	Pole damaged or structurally unsafe
	Hammer	Very hollow sound (all round pole)
	Prodding	Prodding decay to a single depth point of 50mm or about 25mm if all round

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3.2 Routine Inspection and Testing of Wood Poles in Position

a) Visual Inspection

Poles shall be inspected visually from the ground over its entire length above ground for :-

- A sign of decay, such as wet crumbly wood, "D" labels, fungal growth, or evidence that suggests the pole has been "PURL" tested, i.e. nails at ground, 1M and 2M levels.
- Signs of damage or weakness, caused by Woodpeckers, vehicles or farm machinery, excessive animal rubbing. The bending strength of a pole is proportional to the cube of the diameter so particular attention should be paid to external rot or the effect of cattle rubbing on the pole which results in a reduction in the diameter. It is recommended that creosoted laggings be fitted to prevent cattle rubbing where this has occurred previously.
- Signs of splitting or fire damage. Many poles suffer from large cracks due to drying out; these are not detrimental except that they may expose the untreated sapwood and heartwood.
- Damaged insulators or steelwork. If the insulators are damaged the line must be made dead before work is carried out on the support.
- Signs that suggest the pole may not be stable, such as recent excavations around the pole, dikes or trenches nearby, or the gouge mark high above the ground suggesting the pole may not be planted to the correct depth.
- During line inspection a note should be taken of any poles which are buckling. This sometimes occurs on 'light' poles fitted with 'tee off' stays etc. A pole buckling under normal conditions will break during heavy loading and should therefore be programmed for replacement if the buckling is due to load and not just a deformed pole.

b) Hammer Test

Using a 1lb hammer and commencing at ground level hit the pole at close intervals around its circumference, repeat this test at 150mm intervals working up the pole as far as can be reached. Listen for a good ringing sound as the hammer strikes the poles indicating solid timber. A change in tone will be heard when the hammer passes from sound to decayed wood. A dull or dead tone indicates external or slight internal decay whilst a hollow note indicates extensive internal decay. Any areas of wood identified as suspect during the hammer test must be prod tested.

c) Prodding Test

Using a long thin screwdriver or a measuring tool as detailed in Drawing No 1091010606, test the resistance of the timber to prodding by attempting to penetrate the pole surface. This should be done at and just below ground level by using the screwdriver at 45° angle and at areas found to be suspect as a result of the hammer test. Carry out this test at close intervals around the pole circumference. Decayed timber, unlike sound timber, will offer little resistance insertion and removal of the screwdriver blade.

d) Purl Test (Pole Ultra-sonic Rot Locator)

Scan Test

A quick Purl test can be carried out as an addition to the standard hammer test using the Pearl Ultra-sonic tester. This test may be used as a confirmation to the hammer test. Checks shall be carried out at 120° increments around the circumference of the pole at ground level, 500mm, and 1m above ground. Provided no further indications of rot are found no further action is required.

Poles that fail any of the above visual or physical tests must be PURL tested (Full detailed test) to establish the poles soundness, or Supported in accordance with clause 3.8 before being ascended.

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If the tests taken as required by clause 3.2 indicate that the pole is safe to climb, the linesman may climb as required by the work he is to undertake. During his climb attention should be given to the condition of the pole and further hammer tests made if considered necessary.

Pearl Test (Full detailed Test)

Where one of the previous tests (a, b, c or d) indicate the likelihood of rot, a more detailed assessment shall be made with the Pearl tester to assess the degree of residual strength remaining in the pole. To determine the degree of rot the detector shall be placed at nine positions evenly distributed about the circumference of the pole. The results from each test position shall be record on the pre-printed report sheets for later more detailed analysis.

Information on the correct use of the Purl tester can be found in appendix A

3.3 Ground Level Rot

Particular attention to pole rot identification shall be carried out in the area just above ground level. If the tests detailed above indicate the likelihood of decay being present below ground level, then the ground around the pole shall be excavated for a depth of 300 mm, and the section of pole thus exposed tested as above. Observations shall be made to ensure that any changes in previous ground level are identified. Failure to do so could result in unidentified hidden rot.

3.4 Poles suspected of being subjected to shock loading

Poles which are 20 years old and have been subjected to shock loading (i.e. pole or stay being struck by a vehicle, conductors being struck by falling trees or substantial tree limbs, poles where some but not all conductors have snapped and the remaining conductor will be unbound to effect repair) must be approached with caution.

In these circumstances the suspect pole shall be checked by applying leverage by means of a guy rope attached near the top of pole. The guy rope should be attached from the ground by means of J & P rods fitted with a positive grip head attachment and a guy rope applicator as shown on Drawing No 1091450203. The rope should be long enough and the staff pulling on it so placed to preclude any danger should the pole being tests collapse. If, following this test, the pole is still suspected then before it is climbed temporary stays shall be attached. The stays should be attached to the pole by means of the guy rope applicator as shown on Drawing No 1091450203.

If it is not practicable to carry out the check outlined above because of the pole arrangement, i.e. an existing stay. The ground around the base of the pole shall be excavated to a depth of 600 mm and the section of pole thus exposed cleaned and visually inspected. If no signs of fracture are visible, hammer and prod tests should be applied to the exposed area. Only after all three tests have indicated that the pole is sound may it be climbed having first backfilled the excavation at the base of the pole.

Tests which indicate that the pole is unsound must be reported to a supervisor and fitted with a D label as shown on Drawing Number 1091010228.

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3.5 Reinstated Poles (Multi-Tube Pole Reinstatement System)

Recent development work has resulted in a retrofit system whereby existing damaged or rotten poles may be strengthened in situ. This system has the ability to return a decayed or damaged overhead line support to its original strength. Where this system has been applied to an existing pole, the pole shall be deemed to be fit for climbing and use on the network providing that the following additional checks are made: -

- Inspect the Multi-Tube nail for obvious damage or deformation that may have occurred due to vehicular or farm equipment damage. Deformation could also occur due to stress loading following severe weather. If either case is observed the Multi-Tube is recommended for replacement.
- The Multi-Tube is 2400 mm in length with 1200 mm of exposed tube above ground level. A tolerance of +/- 150 mm from the 1200 mm will be accepted due to typical backfill and ground conditions.
- Ensure that no cables or earth wires are obscured or interfered with by the installation of the system.
- Confirm that the Multi-Tube is pulled up as close to the pole as possible
- The Multi-Tube can be fastened to the pole with either M20 bolts or steel straps and a tensioning buckle. When the Multi-Tube is fixed to the pole with bolts ensure that the wood surrounding them is free from rot and that the wood between the top and bottom restraining bolts is sound.

3.6 Notes on rot Assessment

Decay of pole timber is mainly caused by fungal attack at ground level where soil can readily contribute fungi and destructive bacteria. Moisture which is vital for the development of the fungal spores is present in the wood cells for considerably longer periods at ground level than above and hence it is here that the pole will be most intensely attacked.

Pole strength depends not only on the extent of rot, but also on its position in the pole. For example, rot in the centre of a pole with a substantial sound outer ring reduces the strength by a negligible amount. It is sometimes difficult to decide whether rot is sufficiently extensive to necessitate pole changing. In terms of residual strength, this will be provided by the PURL tester.

Minimum levels of residual strength

- >80% Pole may be retained, but shall be treated with Boron Rods.
- <80% Pole shall be replaced.

The results from the PURL tester shall be returned to the supervising engineer to enable residual strength calculations to be carried out. At the time of pole inspection it will be known that a pole has some degree of rot, but the extent and reduction in residual strength will not be known until the calculations have been carried out.

To ensure the safety of our staff, the pole shall be classified as "Unsafe for climbing" and be fitted with a D label as shown on Drawing Number 109101.0228. Once the true remaining strength has been established or the pole has been retrofitted with the multi-tube pole re-instatement system the 'D' label may be removed if appropriate.

3.6.1 Urgency of Pole Replacement

The preliminary works stage of any rebuild program involves a survey of the existing feeder route and the condition assessment of all poles and stays. In some cases this prelim work may well be carried out at least 12 months ahead of any proposed refurbishment work. Inevitably this process will result in the

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discovery of rotten poles and a dilemma exists as to whether the pole should be replaced immediately upon discovery or delayed until the main refurbishment work.

Once the survey data has been returned, the results of the Purl tester can be used to quantify the risk of temporarily leaving the rotten pole on the system.

Pole strength depends not only on the extent of rot, but also on its position in the pole. For example rot in the centre of a pole with a substantial sound outer ring reduces the strength by a negligible amount.

The software supplied with the Purl tester shall be utilised to determine the Minimum Strength remaining in the Pole.

The following figures detail the acceptable limits:

Medium Grade Pole - >70%

Stout Grade Pole - >60%

Provided the minimum strength exceeds the above figures and the pole has been fitted with a 'D' notice as indicated earlier, the pole may be retained and programmed for replacement under the main authorised scheme. This will provide us with a low risk of pole failure and allow a full assessment to be carried out on the feeder as to the correct replacement pole type. (Max period between observation and pole change 2 years)

Where multiple rotten poles are found adjacent to each other, a higher risk to the lines reliability must be present as multiple pole failure represents a substantially increased return to service period under fault conditions.

Where this condition exists and the residual strength of all of the tested poles would have resulted in the poles being allowed to be have been left for delayed replacement a higher value of residual strength of 80% and 70% respectively must be present in all of the effected poles before they may be left for delayed replacement.

If the multiple pole value or the single pole value cannot be met, the poles shall be programmed for replacement as soon as possible. If at this time the information as to the routes up-rating requirement is not known the poles shall be replaced with stout grade poles of the equivalent height as a default.

3.7 Testing Cobra Treated Poles with Aluminium Bandages

Cobra treated poles shall be tested as follows:

- Carry out the Visual, Hammer and prodding tests to exposed pole areas as detailed above.
- Excavate around the pole to just below the bandage (approx 150mm)
- Without removing the bandage, test the pole for indications of rot using the Purl Tester. (The transmitter may be applied to the pole above the aluminium bandage. The receiver shall then be placed at points around the circumference of the pole below the bandage to check for signals)

Provided signals are received in all locations, the bandage can be retained in position.

If no signal is obtained in any location, the bandage shall be removed and the degree of rot diagnosed in the normal way.

After testing, the existing bandage shall be replaced back into its original place.

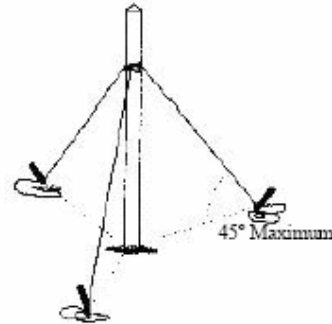
3.8 Making a suspect pole safe to ascend using guy ropes:

- Poles that show minimal signs of damage or decay can be made safe to ascend by securing the pole with guy ropes after which only one person may ascend.
- Using a hydraulic bucket lift, or rope support and live line tapping rods, secure a minimum of three preferably four guy ropes to a point at least two thirds up the pole.

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- Install bars, or ground anchors, a distance of at least the height of the ropes away from the pole. These anchors must be positioned equally around the pole. Remember to check for underground apparatus before knocking in the bars.
- Secure a guy rope to each anchor, take up the tension in all the ropes at the time and tie off securely.

Example using three guy ropes.



3.9 Assessment of steel poles

Corrosion of steel poles is concentrated around ground level, at the joints where the pole changes diameter and at collar positions. Patrolmen are required to pay particular attention to poles which appear bent at the junction of sections or show signs of metal erosion due to rust at or near ground level and report same.

1. Steel poles should be examined for corrosion every 10 years. The cement weather seal should be removed and an ultrasonic thickness test (using an approved tester) carried out to determine the amount of corrosion. It will be necessary to excavate at least 8 cm below ground level. The pole should then be repainted and the area from 8 cm below to 70 cm above ground level painted with two coats of Sovereign K10 polyurethane elastomer. Poles showing corrosion but which do not require changing should be ultrasonically tested at five yearly intervals.

Approved Ultrasonic thickness tester:

Meritronics Digital Ultrasonic Thickness Meter used in conjunction with the UCA-2 couplant.

Visual inspection of the pole will indicate the need for painting. As these poles are usually in villages they should be kept in a reasonable standard of appearance. Repainting should be carried out before the existing coat has completely broken down. The finishing coat may vary depending on local authority requirements.

The table below indicates the original minimum design thickness of the various types of poles. Should the measured thickness be less than 80% of these at any point, then the pole should be replaced if calculations show that the pole is below the required strength.

3.9.1 Ultrasonic testing of Steel Poles.

The following decision sequence shall be used to identify the original steel pole type:

1. Measure base diameter or circumference of pole.
2. Locate base diameter or circumference in table.
3. If only one pole type associated with base size, identify required wall thickness shown on right hand side of table.
4. If more than one pole type indicated, measure length of middle section or bottom section as indicated in the table to identify exact pole type.
5. Use ultrasonic tester to obtain existing wall thickness of pole.

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Table 1

Decision table for the identification of Steel Poles

Base Diameter	Base Circ.	Pole Type	Pole Length	Section Length			Wall Thickness		
				Top	Middle	Bottom	Post 1968	1952 - 68	Pre 1952
168.28	528.7	A		Not Available	Not Available	Not Available	Not Available	Not Available	Not Available
190.5	598.5	B							6.35
193.7	608.6	B	9144 (30 ft)	2134 (7 ft)	2134 (7 ft)	4877 (16 ft)	7.14	8.00	
			9750 (32 ft)	2438 (8 ft)	2438 (8 ft)	4877 (16 ft)	7.94	8.00	
		E	9750 (32 ft)	2850 (9 ft 4 in)	1760 (5 ft 9 in)	5340 (17ft 10in)	10.00		
241.3	758.2	C							7.94
244.5	768.2	C	9144 (30 ft)	2134 (7 ft)	2134 (7 ft)	4877 (16 ft)	9.53	8.00	
			9750 (32 ft)	2438 (8 ft)	2438 (8 ft)	4877 (16 ft)	10.32	9.50	
		F	9750 (32 ft)	2810 (9 ft 3 in)	1910 (6 ft 3 in)	5030 (16 ft 6 in)	12.50		
266.7	838	D							9.53
273.0	857.8	D	9144 (30 ft)	2134 (7 ft)	2134 (7 ft)	4877 (16 ft)	11.10	9.50	
			9750 (32 ft)	2134 (7 ft)	2134 (7 ft)	5486 (18 ft)	11.10	11.10	
		G	9750 (32 ft)	2240 (7 ft 4 in)	2390 (7 ft 10 in)	5120 (16ft 10in)	16.00		
		K	9750 (32 ft)	3000 (9 ft 10 in)	2800 (9 ft 2 in)	4150 (13 ft 7 in)	12.50		

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4. References

4.1 External Documentation

Reference	Title
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4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	31/05/06

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	05/06/06

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	05/06/06

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	21/07/06

6.5 Approval

Approval is given for the content of this document

		Sign	Date
D Sillito	Major Project Manager South	D Sillito	17/06/06
S Gardner	OHL Construction Manager	S Gardner	16/06/06

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
G Earl	Primary Engineering Projects Manager	G Earl	19/06/06

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PURL Testing Wood Poles

Background

A PURL uses an ultrasonic signal to identify the presence and locate the position of decayed timber in wood poles. The transmitter of the PURL is fixed to the pole and when turned on gives out a ultrasonic signal that can only travel through sound timber. The receiver is then held against the pole at the various positions to test for the transmitter signal. When turned on the light on the receiver should flash in unison with the transmitter "clicks". If there is decayed timber between the transmitter and the receiver then there will be an out of unison or "No Signal".

General :-

- Visually inspect the pole for signs of damage such as decay fungal growth, splitting, etc... or weakness caused by woodpeckers, vehicles, farm machinery, excessive animal rubbing, etc...
- Clear away any raised ground at the base of the pole to reveal the true ground level.
- Before starting work check that the PURL testing kit contains one of each of the following items of equipment ; -
Transmitter, Receiver, Ratchet drilling device, Drill 7/64", Measuring Tape, "D" marker plate, Calibration block (Go), Calibration block (No Go), Elastic marker band (small poles), Elastic marker band (large poles). Results form YE 1190 4/82 and the Carrying case.
- When turned on and applied to a pole the light on the receiver should flash in unison with the transmitter "clicks". If there is decayed timber between the transmitter and the receiver then there will be no flashes or the flashes will be out of unison with the clicks from the transmitter.
- Complete a "Scan test" first to determine the condition of the pole. If you fail to get a signal or you get a signal which is out of unison you must complete a "Full test" and record the results using form YE 1190 4/82 as shown on page 4.
- The transmitter and receiver must be checked for correct operation before and after use.

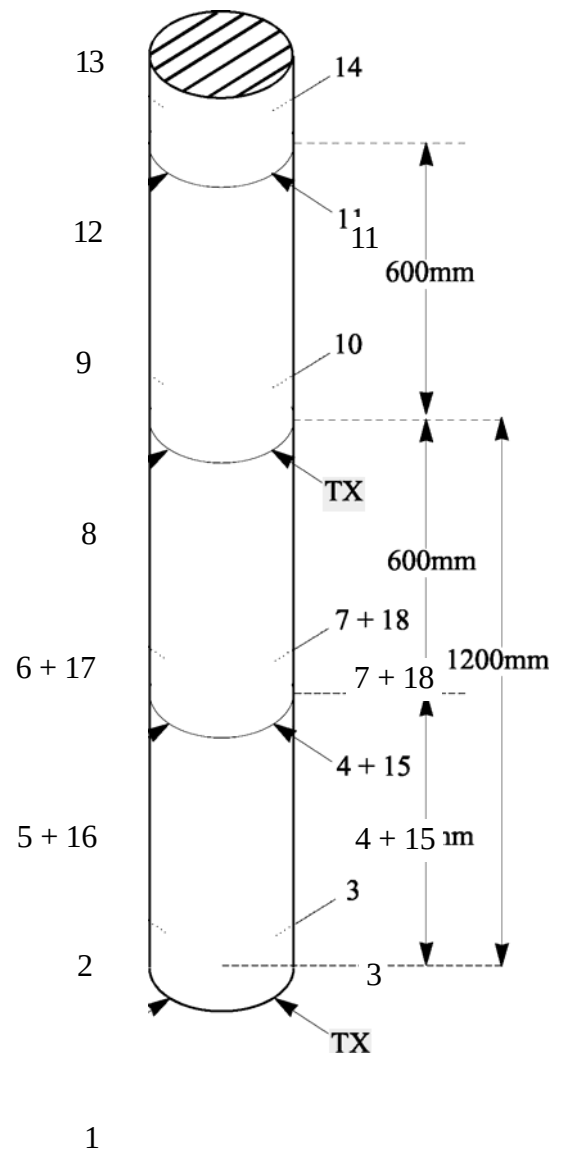
Checking the PURL Equipment

- Stand the PURL on its end with the screw uppermost and screw on the "Go" test block.
- Switch on the transmitter by turning the control switch clockwise. A series of "clicks" should be heard.
- Switch on the receiver to position 1 and apply the receiver to the test block at the opposite end to the transmitter, a signal should be obtained.
- Unscrew the "Go" test block and screw on the "No go" test block. Apply the receiver to the test block, a signal should not be obtained.
- If either the transmitter or receiver fails to operate the batteries may need changing. Access to the batteries is obtained by unscrewing the end cap. If either device still fails to operate correctly return them to your supervisor.

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Scan Test

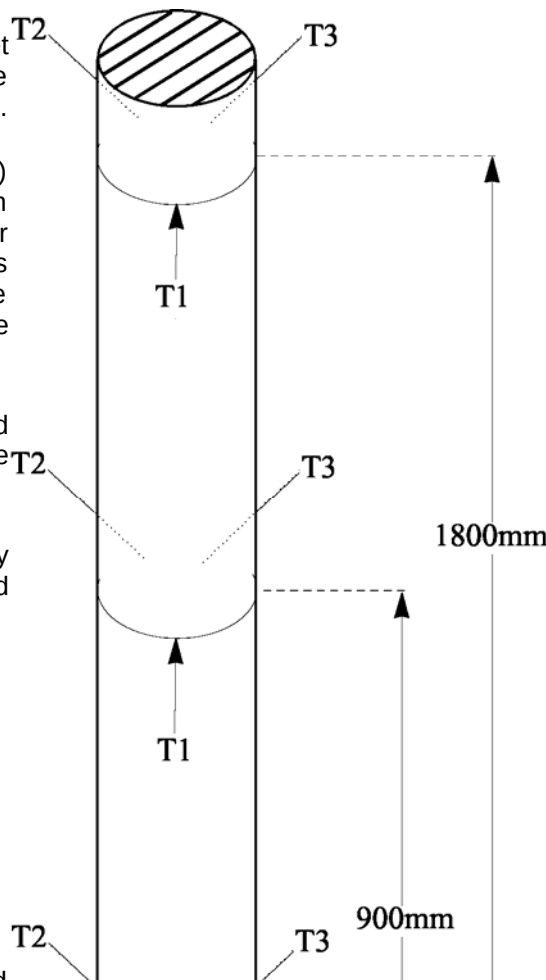
1. Drill a pilot hole in the pole at ground level and in line with one set of conductors using the ratchet drill. Screw the transmitter into the pilot hole to the full extent of the thread and turn back half a turn. Turn on the transmitter.
2. Switch on the receiver by turning the control switch to the required sensitivity setting. The setting is determined by measuring the diameter of the pole using the tape measure at a height of 1 m from the ground. The tape gives a reading of the setting to be used. Care must be taken if there are cables fixed to the pole.
3. Apply the receiver at points 1, 2 and 3 shown on the illustration.
4. Leave the transmitter in place and test using the receiver at 600mm above ground level at points 4, 5, 6 and 7 shown in the illustration.
5. Re-position the transmitter directly above the first position at 1200mm above ground level. Test using the receiver at points 8, 9 and 10 shown on the illustration.
6. Leave the transmitter in place and test using the receiver at 600mm above the transmitter at points 11, 12, 13 and 14 shown in the illustration.
7. Leave the transmitter in place and test using the receiver at 600mm below the transmitter at points 15, 16, 17 and 18 shown in the illustration.



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Full Test

1. Fix the transmitter to the pole, in line with one set of conductors at a height of 1800mm from the ground. This is the Transmitter "T1" position.
2. At the same level but 120 degrees (1/3rd) around the pole in a clockwise direction around the pole chalk mark the Transmitter "T2" position and at a further 120 degrees (1/3rd) around the pole in a clockwise direction around the pole chalk mark the Transmitter "T3" position.
3. Fix the appropriate size elastic marker band around the pole at the same height as the transmitter.
4. Using the receiver, test for signals at 9 equally spaced positions as indicated by the marker band moving in a clockwise direction around the pole.
5. Move the transmitter to the T2 position and repeat step 4.
6. Move the transmitter to the T3 position and repeat step 4 again.
7. Repeat steps 1 to 6 with the transmitter at 900mm from ground level.
8. Repeat steps 1 to 6 with the transmitter at ground level.
9. Permanently mark all T1 positions using marker nails. Use 1 nail at ground level, 2 at 900mm and 3 at 1800mm above ground level.



Recording and interpreting results

- Fill in a "Wood Pole Test Results" form YE 11904/82 for each pole on which you complete a Full Scan. A sample of a completed Wood Pole Test Results form is shown overleaf.
- The receiver positions are numbered 1-9 from the present transmitter position moving clockwise around the pole.
- Enter "X" for every position where you fail to get a signal or receive an out of unison signal.
- Enter "-" for any position where the receiver cannot be applied because of an obstruction or damage to the pole.
- If any of the boxes have 10 or more crosses then a red "D" marker plate must be fixed to the pole. Using brass screws, fix the marker plate to the pole under the number plate at the highest point that you can reached from ground level.

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Details of support tested

WOOD POLE TEST RESULTS

location: SCARCROFT

district: 41 line no: A721 part: 01 support no: 45

height of test 1 (mm): Ø

1	2	3	4	5	6	7	8	9	
X	X	X			X	X			T1
			X	X	X		-	-	T2
		X	X	X	X				T3

height of test 2 (mm): 9,00

1	2	3	4	5	6	7	8	9	
									T1
									T2
									T3

height of test 3 (mm): 18,00

1	2	3	4	5	6	7	8	9	
									T1
									T2
									T3

date of test: 170895

view from top of pole

A. Linesman
signature

YE 1130 4/02



pole leaning
yes ☐

pole split
yes ☐

pole burnt
yes ☐

"D" marker left
yes ☒

comments
S2

"X" indicates no signal or out of unison signal at this receiver position

Tick "D": if marker plate fitted

Difficult Situations

Where a pole is situated close up to an obstruction it will normally be possible to select 3 approximately equally spaced positions for the transmitter. In these situations the T1 position does not have to be in line with the conductors if it is not possible. The receiver must be used in as many of the positions as it is physically possible to reach.